

Q1

$$\sim (p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$$

Proof. $\sim (p \vee \sim q) \vee (\sim p \wedge \sim q)$

$$\begin{aligned} &\equiv (\sim p \wedge q) \vee (\sim p \wedge \sim q) && \text{(by De Morgan laws)} \\ &\equiv \sim p \wedge (q \vee \sim q) && \text{(by distributive law for } \wedge \text{)} \\ &\equiv \sim p \wedge \mathbf{t} && \text{(by negation law for } \vee \text{)} \\ &\equiv \sim p && \text{(by identity law for } \wedge \text{)} \end{aligned}$$

Q2

$$p \vee \sim q \rightarrow r \vee q$$

Proof. (a) $p \vee \sim q \rightarrow r \vee q \equiv \sim (p \vee \sim q) \vee (r \vee q)$

Q3

Determine whether the statements in (a) and (b) are logically equivalent.

(a) Bob is both a math and computer science major and Ann is a math major, but Ann is not both a math and computer science major.

(b) It is not the case that both Bob and Ann are both math and computer science majors, but it is the case that Ann is a math major and Bob is both a math and computer science major.

Proof. Define

1. p : Bob is a math major.
2. q : Bob is a CS major.
3. r : Ann is a math major.
4. s : Ann is a CS major.

The statement in (a) has the form: $(p \wedge q \wedge r) \wedge \sim (r \wedge s)$

The statement in (b) has the form: $\sim (p \wedge q \wedge r \wedge s) \wedge (r \wedge p \wedge q)$

We could write down the entire truth table with $2^4 = 16$ rows, but let's try to take some shortcuts.

If we look at the two expressions, they both have the form $(p \wedge q \wedge r) \wedge Z$. Such an expression is true when $(p \wedge q \wedge r)$ and Z are both true, and it is false otherwise.

Therefore, when evaluating Z , we can concentrate only on the rows of the truth table where $(p \wedge q \wedge r)$ is true. We can disregard Z on the other rows (because both (a) and (b) are false for those rows).

$(p \wedge q \wedge r)$ is true only when all 3 of p, q, r are true. When this is the case, we only need to consider the possibilities for s . This gives us 2 rows only:

s	$\sim (r \wedge s)$	$\sim (p \wedge q \wedge r \wedge s)$
T	F	F
F	T	T

The two Z expressions have the same truth values in all cases. Therefore (a) and (b) are logically equivalent. $\square$